VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY
HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY
**FACULTY OF COMPUTER SCIENCE AND ENGINEERING**

Multidisciplinary Project

# Midterm Report

**Lecturer:** Nguyen Thanh Loc
**Class:** CC05
**Group:**

**Group members:**

| | |
|---|---|
| Le Dang Khanh Quynh | 2353037 |
| Tao Nguyen Quang Khang | 2353212 |
| Vu Quoc Viet | 2353326 |
| Le Nguyen Tuan Khoi | 2450014 |

**Group contribution:**

| No. | Fullname | Student ID | Percentage of work |
|---|---|---|---|
| 1 | Dong Gia Bao | 2352770 | 20% |
| 2 | Tao Nguyen Quang Khang | 2353037 | 20% |
| 3 | Vu Song Anh | | 20% |
| 4 | Vu Quoc Viet | 2353326 | 20% |

# Contents

**1 Introduction** **2**
1.1 System Description . . . 2

**2 Input and Output Devices** **2**
2.1 Input Devices . . . 2
2.2 Output Devices . . . 2

**3 User Requirements** **2**
3.1 Functional Requirements . . . 2
3.2 Non-Functional Requirements . . . 2

**4 Use Case Detail** **3**

**5 System Architecture** **4**
5.1 Deployment View . . . 4

**6 Data Flow** **4**

**7 Module Breakdown** **5**

**8 Hardware Plan** **5**
8.1 Flame Detection Modules . . . 5
8.2 Smoke and Gas Detection Modules . . . 5
8.3 Temperature Sensors . . . 5
8.4 Microcontrollers . . . 6
8.5 Alarm and Output Modules . . . 6
8.6 Control Modules . . . 6
8.7 Communication Modules (IoT) . . . 6
8.8 Optional Modules . . . 6

# 1 Introduction

## 1.1 System Description

The Fire Alarm System is designed to detect early signs of fire such as smoke, abnormal gas level, and high temperature. The system collects sensor data, sends them to the server, and warns users through local alarm devices and an application.

# 2 Input and Output Devices

## 2.1 Input Devices

- Smoke sensor
- Gas sensor
- Temperature sensor
- Emergency button

## 2.2 Output Devices

- Buzzer
- LED warning light
- Mobile/Web notification

# 3 User Requirements

## 3.1 Functional Requirements

- The system can monitor sensor data in real time.
- The system can detect fire risk conditions.
- The system can activate alarms when danger is detected.
- The system can notify users through the application.

## 3.2 Non-Functional Requirements

- The system should respond in real time.
- The system should operate continuously and reliably.
- The application should be easy to use.

# 4 Use Case Detail

| | |
|---|---|
| Use Case ID | UC01 |
| Use Case Name | Monitor sensor data |
| Actors | User, Hardware Devices |
| Description | The system collects data from sensors and displays them on the application. |
| Preconditions | Sensors are connected and working properly. |
| Normal Flow | 1. Sensors collect data.<br>2. Controller sends data to server.<br>3. Server stores and processes data.<br>4. Application displays current data. |
| Alternative Flow | If one sensor is unavailable, the system continues monitoring available sensors. |
| Exceptions | If the server connection fails, the application cannot update the latest data. |

Table 1: Use case detail for monitoring sensor data

# 5 System Architecture

## 5.1 Deployment View

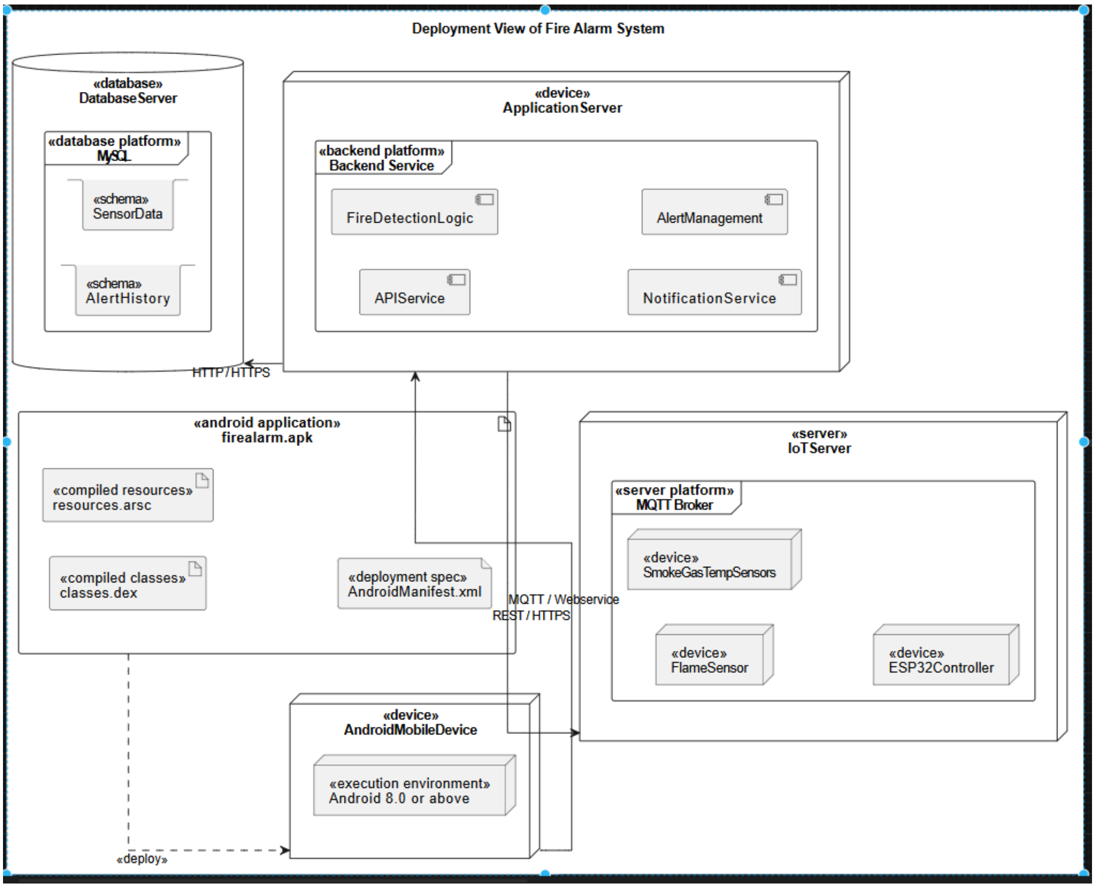


Figure 1: System deployment view of Fire Alarm System

The system consists of hardware, server, and application layers. Sensor devices send environmental data to the controller, which forwards them to the server. The server processes data, stores information, and sends warnings to the application.

# 6 Data Flow

Sensor data are collected from the hardware layer and transmitted to the server through the controller. The server analyzes the data and determines whether the environment is safe or dangerous. Then, the processed results are stored in the database and sent to the application. If a fire risk is detected, the system triggers buzzer and LED alarms while also sending warning notifications to users.

# 7 Module Breakdown

| Module | Description |
|---|---|
| Sensor Module | Collects smoke, gas, and temperature data |
| Controller Module | Reads sensors and sends data to server |
| Alarm Module | Activates buzzer and LED warning devices |
| Server Module | Processes data and stores system status |
| Application Module | Displays system status and notifications |

Table 2: Main modules of the system

# 8 Hardware Plan

The Fire Alarm System utilizes several hardware components for sensing, processing, communication, and alerting. The main hardware modules are described as follows:

## 8.1 Flame Detection Modules

- **KY-026 Flame Sensor Module**
  - Detects infrared light emitted by flames to identify fire presence.
- **YG1006 IR Flame Sensor Module**
  - Senses IR radiation from fire sources and outputs digital/analog signals.

## 8.2 Smoke and Gas Detection Modules

- **MQ-2 Gas Sensor Module**
  - Detects smoke and flammable gases such as LPG, methane, and hydrogen.
- **MQ-135 Gas Sensor Module**
  - Measures air quality and detects gases like CO2, NH3, and benzene.

## 8.3 Temperature Sensors

- **DHT11 Temperature and Humidity Sensor**
  - Measures ambient temperature and humidity for environmental monitoring.
- **DS18B20 Temperature Sensor**
  - Provides precise digital temperature readings for detecting abnormal heat.

## 8.4 Microcontrollers

- **Arduino Uno R3**
  - Processes sensor data and executes control logic.
- **ESP32 DevKit V1**
  - Handles data processing and enables wireless communication via WiFi/Bluetooth.

## 8.5 Alarm and Output Modules

- **KY-012 Active Buzzer Module**
  - Generates an audible alarm when fire is detected.
- **12V Piezo Siren Alarm**
  - Produces a loud emergency alert sound.

## 8.6 Control Modules

- **SRD-05VDC-SL-C Relay Module**
  - Controls high-power devices such as sirens or electrical circuits.

## 8.7 Communication Modules (IoT)

- **ESP8266 ESP-01 WiFi Module**
  - Enables wireless data transmission to servers or cloud systems.
- **SIM800L GSM/GPRS Module**
  - Sends SMS or data alerts in emergency situations.

## 8.8 Optional Modules

- **DS3231 RTC Module**
  - Maintains accurate time for event logging.
- **MicroSD Card Module**
  - Stores sensor data locally for backup or analysis.